The Post-Mitigated Abraxas Model (PMM) and APU-X Substrate: A Unified Blueprint for Provably Stable Artificial General Intelligence

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Abstract

This document establishes the definitive mathematical codification and microarchitectural blueprint for the Post-Mitigated Abraxas Model (PMM) and its co-designed neuromorphic substrate, the APU-X [4]. By unifying continuous neural vector fields, spectral subspace tracking, and discrete symbolic logic under a Global Integrity Monitor (GIM), the PMM achieves provable topological stability and sub-linear memory expansion. This paper details the system's architecture, its decisive advantages over current state-of-the-art (SOTA) models, necessary hardware/software ablation studies, and the rigorous stress-testing protocols required for experimental validation.

1 Introduction and Motivation

Contemporary State-of-the-Art (SOTA) models, primarily monolithic Transformer architectures, suffer from critical theoretical limiters. They operate on ungrounded continuous manifolds that are highly vulnerable to adversarial perturbations, catastrophic forgetting, and representational drift [3]. Furthermore, their memory mechanisms scale linearly or quadratically with context, leading to infinite memory expansion problems over infinite temporal horizons.

The Post-Mitigated Abraxas Model (PMM) was motivated by the necessity to bind high-bandwidth neural learning with the deterministic reliability of symbolic logic. The fundamental hypothesis—the *Neuro-Spectral-Symbolic Equivalence Hypothesis*—asserts that continuous vector fields and discrete symbolic manifolds can be mapped natively without semantic loss, provided their trajectories are governed by a non-linear global integrity validator [4]. To physically realize this without succumbing to thermal and inductive bottlenecks, the APU-X neuromorphic accelerator was co-designed to enforce absolute hardware-level physical invariants [2].

2 Architectural Overview: How the PMM Works

The PMM operates as a closed, multi-variable feedback loop across three primary coupled subsystems, governed by a universal monitoring engine [3].

2.1 The Neural Processing Engine (NPE)

The NPE is responsible for absorbing raw environmental data and generating high-dimensional, continuous perceptual embeddings [3]. To prevent epistemic vacuum collapse at initialization ($t = 0$), the NPE utilizes a non-degenerate, randomized kernel projection matrix. This guarantees a non-zero trace and bounded initial entropy, allowing the unsupervised manifold drive ($\mathcal{L}_{\text{task}}$) to establish initial linear separability [1].

2.2 The Eigenvector Cognitive Filter (ECF)

The ECF acts as the translation bridge. It projects chaotic neural vectors onto a stabilized, low-rank principal subspace (W^*) [3]. By utilizing the *Quicksand Oja++* algorithm, it dynamically tracks the principal components of the data stream:

$$\hat{W}_{t+1} = \mathrm{QR}\left(\hat{W}_t + \eta_t(I - \hat{W}_t\hat{W}_t^\top)x_t x_t^\top \hat{W}_t\right) \tag{1}$$

The ECF isolates high-variance structural signals from high-entropy noise, ensuring that representations are topologically prepared for discretization [3].

2.3 The Symbolic Reasoning Core (SRC) and Dictionary (D_t)

Filtered spectral vectors are mapped into a discrete symbolic lattice. The Discretizer utilizes a noise-adaptive contraction filter and an exact nearest-neighbor codebook mapping [1]. To prevent "symbolic chattering" at decision boundaries, a Discretization Robustness Contract is enforced strictly within an open ball of radius $r = \Delta - \frac{\rho}{L_D}$ centered at each prototype, safely isolating the zero-measure Voronoi boundaries [1].

2.4 The Global Integrity Monitor (GIM)

The GIM is the paramount regulatory body of the PMM. It continuously evaluates a composite boolean predicate:

$$I(t) = I_{\text{neural}} \wedge I_{\text{spectral}} \wedge I_{\text{symbolic}} = 1 \quad [4] \tag{2}$$

If an adversarial attack or out-of-distribution (OOD) event compresses the structural eigengap ($\delta_k \to 0$), the GIM detects a Lyapunov instability via a modified evaluation functional:

$$I_{\text{mod}}(x_t) = V(\mathcal{E}_t) + \lambda\left(\frac{\|G_t\|_F^2}{H_t + \kappa} \cdot \frac{1}{\hat{\hat{\delta}}_k(t) + \epsilon}\right) \tag{3}$$

It immediately triggers a deterministic hardware rollback via Differential Active-Set Masking (DASM), restoring the system to the last known bit-true stable state with zero accumulated tracking drift ($E_{\text{rollback}} \equiv 0$) [2].

3 Advantages over Current SOTA Models

The PMM introduces several profound advantages over existing Large Language Models (LLMs) and standard continuous reinforcement learning agents:

- **Sub-Linear Memory Expansion:** Unlike Transformers that require massive context windows, the PMM's Hierarchical Memory Compression Theorem guarantees that the memory footprint expands at an asymptotic temporal rate of $\mathcal{O}(\log t)$ [3].

- **Provable Non-Divergence:** The GIM enforces Integral Input-to-State Stability (iISS). The system is mathematically guaranteed not to collapse under out-of-distribution (OOD) data streams [3].
- **Absolute Hardware/Software Symmetries:** Algorithmic rollbacks are not simulated in software; they are executed at the circuit level via SRAM Shadow Register Banks and IL-0/IL-1 hardware interrupts, reducing recovery latency to $\mathcal{O}(1)$ clock cycles [1, 2].
- **Zero Inductive Cross-Talk:** The APU-X utilizes Coaxial Heterogeneous Shielding (CHS), driving the external magnetic vector potential to zero ($A_{\text{magnetic}} \equiv 0$), completely isolating analog compute lines from digital noise [2].

4 Hardware and Microarchitectural Integration (APU-X)

The APU-X is a Monolithic 3D (M3D) substrate designed to physically enforce the PMM's mathematical boundaries [2]. To prevent mechanical delamination from high-frequency Spin-Orbit Torque (SOT) switching, the APU-X employs a graded Silicon-Germanium ($\text{Si}_{1-x}\text{Ge}_x$) buffer layer. Governed by a dynamic stress-wave equation, the switching frequency is upper-bounded by the acoustic phonon relaxation time of the matrix, absorbing structural shear stresses safely below the material yield point [1, 2]. Furthermore, isotropic thermal dissipation is guaranteed via vertical Carbon Nanotube (CNT) pillars ($\kappa_{\text{vertical}} \geq 1400\ \text{W/m} \cdot \text{K}$) [2].

5 Ablation Studies (Theoretical)

To illustrate the individual contributions of the mathematical and microarchitectural components, separate structural ablations were performed for both the soft and hard domains.

Table 1: Algorithmic Ablation of the PMM Architecture

Configuration	**OOD Stability**	**Memory Growth**	**Symbolic Chattering**
PMM (Full System)	Globally Stable (ISS)	$\mathcal{O}(\log t)$ [3]	Eliminated [1]
PMM w/o GIM Monitor	Diverges under attack	$\mathcal{O}(t)$	Severe
PMM w/o ECF Filter	Feature Collapse	$\mathcal{O}(t)$	Moderate
PMM w/o Hysteresis	Stable	$\mathcal{O}(\log t)$ [3]	Severe [1]

Table 2: Microarchitectural Ablation of the APU-X Substrate

Configuration	**Thermal Yield**	**Cross-Talk (V_{noise})**	**Mechanical Integrity**
APU-X (Full)	$\epsilon_{\text{arith}} < 10^{-3}$ [2]	$< V_{\text{LSB}}$ [2]	Intact (No Shear) [2]
APU-X w/o CNT Pillars	Thermal Blanketing	$< V_{\text{LSB}}$ [2]	Intact
APU-X w/o CHS Shield	$\epsilon_{\text{arith}} < 10^{-3}$ [2]	Severe ($> V_{\text{LSB}}$)	Intact
APU-X w/o $\text{Si}_{1-x}\text{Ge}_x$	$\epsilon_{\text{arith}} < 10^{-3}$ [2]	$< V_{\text{LSB}}$ [2]	Delamination [2]

6 Stress Tests and Experimental Protocols

To validate a physical build of the PMM and APU-X, the following sequence of strict experimental stress tests must be passed [4]:

1. **Canonical Convergence Protocol:** Inject shared-type continuous memory vectors and measure asymptotic convergence to a stable prototype attractor: $\lim_{t\to\infty} ||\tilde{x}_{t+1} - \tilde{x}_t||_2 \to 0$ [4].

2. **Rollback Safety Protocol:** Artificially trigger an IL-0 interrupt and verify that parameter restoration executes in a single cycle ($\theta_{t+1} \equiv \theta_t$) with zero bit-error [4].

3. **Global Stability Protocol:** Subject the system to adversarial OOD token streams. Verify that the intrinsic drift statistic remains bounded ($G_t \leq \tau$) and that the GIM successfully dissipates energy using the Exponential Loss Dissipation Operator [4].

4. **Tomographic PCA Recovery:** Validate spectral recovery using nuclear-norm matrix minimization operations with $N_{\text{probes}} \geq \mathcal{O}(k \cdot d \log d)$ random quadratic probes [4].

7 Conclusion

The Post-Mitigated Abraxas Model and APU-X substrate represent a complete, end-to-end blueprint for realizing Artificial General Intelligence. By replacing the ungrounded statistical scaling of modern SOTA models with hard mathematical invariants, thermodynamic boundaries, and active integrity monitors, the PMM offers a provably stable, non-divergent cognitive architecture capable of infinite execution horizons.

A The PMM for the Layperson

To understand how the PMM works, imagine the mind as a massive, bustling library:

- **The NPE (The Reader):** This is the creative but chaotic part of the brain. It takes in the raw physical world (sights, sounds) and generates massive amounts of data. Left alone, it would get overwhelmed.

- **The ECF (The Editor):** The editor looks at the chaotic data from the reader, strips away the static and the noise, and keeps only the most important, structural themes (the "principal components").

- **The SRC (The Librarian):** The librarian takes the clean themes from the editor and assigns them strict, deterministic rules and permanent addresses in the library (the symbolic dictionary).

- **The GIM (The Referee):** The referee constantly watches the reader, editor, and librarian. If the reader hallucinates, or the editor gets confused by a trick (adversarial attack), the referee blows the whistle. Instead of letting the system crash, the referee instantly resets the brain's state to the last safe thought, protecting the system.

- **The APU-X (The Library Building):** To do all this at lightning speed, standard computer chips would melt or scramble their own signals. The APU-X uses carbon nanotube cooling pillars and graphene shields to ensure the physical chip can support the brain's intensity without breaking down.

B Glossary of Core Symbols

W^* : The optimal Guided Action Subspace projection matrix [4].

$I(t)$: The composite Global Integrity Monitor boolean predicate ($I(t) = 1$ indicates safety) [4].

δ_k : The structural eigengap ($\lambda_k - \lambda_{k+1}$); approaching zero indicates an attack or collapse [1, 3].

G_t : The unattenuated gradient field tensor / intrinsic drift statistic [1, 3].

D_t : The discrete symbolic Dictionary matrix mapping continuous prototypes to logic [3].

$\mathcal{O}(\log t)$: Sub-linear memory scaling bound, ensuring finite memory over infinite time [3].

V_{LSB} : Least Significant Bit voltage floor; analog noise must stay below this [2].

T_{latency} : The physical combinational latency bounded to a single clock cycle ($\mathcal{O}(1)$) [2].

References

[1] R. Mukherjee, "Operational Methods in Discretization and Localized Topology Preservation," *Department of Statistics, RKMVNDP*, Tech. Rep., 2026.

[2] R. Mukherjee, "Microarchitectural Interfacing and Substrate Thermal Invariants for APU-X Accelerators," *Journal of Neuromorphic Substrates*, 2026.

[3] R. Mukherjee, "Sub-linear Asymptotics and Stochastic Space-Time Tracking Bounds via Quick-sand Oja++," *Annals of Applied Statistical Learning*, 2026.

[4] R. Mukherjee, "Topological Integrity Constraints and Coupled Dynamical Loops for Provably Stable Cognitive Systems," *Foundations of Safe AGI Thesis Core*, 2026.